Ep 12 · KYC & Fraud Copilot — Reference Answer on One Page

AI System Design · AI Engineering with Ajit · github.com/byruajit/ai-system-design

SHAPE OF THE SYSTEM

Capture → **parallel checks** (OCR/doc · face/liveness · sanctions/PEP · device/graph) → features → **fraud model (classic ML)** → **rules** → auto-approve **or** analyst case → **LLM copilot** (read-only summary) → **analyst decides** → audit

ROLES

Model scores (calibrated, reason codes) · **LLM explains** (grounded summary, policy citations) · **Human decides** (flagged cases)

NUMBERS (NORTHWIND)

15k apps/day · burst ~2/s · 60% straight-through · 6k cases/day · 12 → 5 min/case · ~$0.004 LLM/case *(illustrative)* · ~45 GB images/day

MUST-SAY POINTS

- Why not an LLM for scoring: calibration, validation, cost, latency
- Any potential sanctions hit → human
- Vendor down → fallback or analyst queue, **never auto-reject**
- OCR text is untrusted (injection)
- Audit: checks · score · reasons · versions · decision
- Residency: India and US stacks separate

EVALS

Risk model: AUC · calibration · FPR by segment · PSI drift · champion–challenger Copilot: faithfulness (numbers ∈ facts) · analyst ratings · time/case Fairness: no protected attributes; monitor outcomes

THE 9 STEPS, ONE MORE TIME

Clarify · Requirements · Estimates · HLD · Deep dive · Data flow · Scale/reliability/security · Evals · Trade-offs

Hire bar: rubric average ≥ 7, no dimension below 5

Northwind Bank is fictional. Prices are illustrative tiers — check current provider pricing.